

Survival Benefit Associated With the Use of Extracorporeal Life Support for Neonates With Congenital Diaphragmatic Hernia

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Objective: To measure the survival among comparable neonates with CDH supported with and without ECLS.

Summary of Background Data: Despite widespread use in the management of newborns with CDH, ECLS has not been consistently associated with improved survival.

Methods: A retrospective cohort study was performed using ECLS-eligible CDH Study Group registry patients born between 2007 and 2019. The primary outcome was in-hospital mortality. Neonates who did and did not receive ECLS were matched based on variables affecting risk for the primary outcome. Iterative propensity score-matched, survival (Cox regression and Kaplan-Meier), and center effects analyses were performed to examine the association of ECLS use and mortality.

Results: Of 5855 ECLS-eligible CDH patients, 1701 (29.1%) received ECLS. “High-risk” patients were best defined as those with a lowest achievable first-day arterial partial pressure of CO₂ of ≥ 60 mm Hg. After propensity score matching, mortality was higher with ECLS (47.8% vs 21.8%, odds ratio 3.3, 95% confidence interval 2.7–4.0, hazard ratio 2.3, $P < 0.0001$). For the subgroup of high-risk patients, there was lower mortality observed with ECLS (64.2% vs 84.4%, odds ratio 0.33, 95% confidence interval 0.17–0.65, hazard ratio 0.33, $P = 0.001$). This survival advantage was persistent using multiple matching approaches. However, this ECLS survival advantage was found to occur primarily at high CDH volume centers that offer frequent ECLS for the high-risk subgroup.

Conclusions: Use of ECLS is associated with excess mortality for low- and intermediate-risk neonates with CDH. It is associated with a significant survival advantage among high-risk infants, and this advantage is strongly influenced by center CDH volume and ECLS experience.

Keywords: CDH, congenital diaphragmatic hernia, ECLS, ECMO, extracorporeal life support, extracorporeal membrane oxygenation

(Ann Surg 2020;xx:xxx–xxx)

Congenital diaphragmatic hernia (CDH) is one of the most complex and resource-intensive anomalies managed by pediatric and neonatal specialists.^{1–3} In the management of CDH, multidisciplinary teams, including maternal-fetal medicine specialists,

neonatologists, intensivists, pediatric surgeons, and a myriad of other subspecialists, collaborate and use diagnostic imaging, mechanical ventilation, pharmacotherapy, surgery, and extracorporeal life support (ECLS) to optimize outcomes. Approximately one-third of infants with CDH – those with the highest-risk disease including more severe pulmonary hypertension,⁴ pulmonary hypoplasia,⁵ and cardiac dysfunction⁶ – receive ECLS as an early, post-natal, life-sustaining intervention.

The efficacy of ECLS for neonates and children with respiratory failure is a topic of ongoing controversy.⁷ CDH is the most common diagnosis among neonates with respiratory failure receiving ECLS,^{8,9} and use of ECLS has been associated with higher survival rates for neonates with severe CDH.^{10–13} However, CDH severity is variable,¹⁴ as are center criteria for ECLS use,¹⁵ center ECLS and CDH volume,^{16,17} and individual clinician or team experience and bias, all of which potentially affect outcomes with this modality. Operative and management complications, resource utilization, and long-term sequelae must also be considered when assessing the utility of ECLS for CDH. Therefore, the answer to the question of whether ECLS is truly beneficial in CDH is complex and remains elusive.

The purpose of this study was to quantify the association between ECLS use and mortality in CDH. Moreover, we sought to measure the influence of center CDH volume and ECLS experience on survival in ECLS-treated patients. We hypothesized that ECLS would be associated with improved survival for patients with high-risk CDH, and that the best survival would be seen at centers that have not only the highest CDH and ECLS volumes (the most experience), but also have high rates of ECLS use for high-risk patients and low rates of ECLS use for low-risk patients (optimal patient management and selection).

METHODS

Study Population

This is a 12-year retrospective study using the CDH Study Group, (CDHSG; Supplemental Digital Content; <http://links.lww.com/SLA/C163>) registry (birth dates 2007–2019), which is a prospectively maintained multinational voluntary database of CDH patients. Clinical variables span the time period from prenatal diagnosis through primary hospital discharge or death. Research using the registry is approved by the University of Texas McGovern Medical School in Houston Center for the Protection of Human Subjects/Institutional Review Board (#HSC-MS-03-223; Ref #118886).

This analysis included only patients meeting ECLS eligibility criteria, specifically those with an estimated gestational age at birth ≥ 32 weeks and a birth weight ≥ 1.8 kg with no severe life-threatening associated cardiac or chromosomal anomalies.

The primary outcome of in-hospital mortality was compared between 2 groups: patients who were treated with ECLS [(+)ECLS], and patients that were not [(-)ECLS]. Patients who survived to discharge from hospital were considered survivors.

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The authors report no conflicts of interest.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal’s Web site (www.annalsofsurgery.com).

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ISSN: 0003-4932/16/XXXX-0001

DOI: 10.1097/SLA.0000000000003928

Unadjusted Analysis and Risk Stratification

Standard univariate comparison between the 2 unadjusted groups was performed, and multivariable logistic regression was used to identify optimal independent predictors of mortality that could be used for risk stratification. Patient severity subgroups were generated using 2 variables: (1) the lowest partial pressure of carbon dioxide (PCO₂) during the first 24 hours of life (“lowest PCO₂”), and (2) the composite probability of ECLS treatment or death without ECLS (“ECLS probability”), using an established 4-covariate risk stratification model (1 and 5 minute Apgar scores and the highest and lowest PCO₂ during the first 24 hours of life).^{18–24} The ECLS probability model was used because it was developed from the CDHSG registry to compare risk-adjusted ECLS usage between centers.^{22,25}

Threshold values of lowest PCO₂ and ECLS probability were identified beyond which a significant survival benefit was seen in the (+)ECLS group. These thresholds were tested after propensity score (PS) matching, then if confirmed, were used to stratify patients into risk categories for subsequent analysis.

PS Matching

ECLS-eligible patients were matched using PSs derived from a multivariable logistic regression model, with ECLS treatment as the dependent variable. All potentially confounding covariates were included in the development of the PS model; however, only data reasonably available at delivery or during the early postnatal period were used, to approximate the setting of a randomized trial. These included 12 variables, many of which have been previously associated with mortality and/or use of ECLS: sex, race,²⁶ prenatal diagnosis,²⁷ gestational age at birth (estimated gestational age) and birth weight,²⁸ high or low CDH volume center,²⁹ inborn or outborn status,³⁰ CDH laterality,³¹ 1 minute and 5 minutes Apgar scores,³² and highest and lowest partial PCO₂ during the first 24 hours of life.^{18,24,33}

Three separate PS matches were performed. Covariate balance pre- and post-matching was assessed using Pearson Chi-squared, Fisher exact, or Student *t* test with balance defined as $P \geq 0.05$.

Match 1: Total Cohort

All (+)ECLS patients were PS matched 1:1 to (–)ECLS patients. A radius matching algorithm was used that optimizes close matches, with a maximum PS radius of 1%. The odds ratio (OR) for death with ECLS use was calculated in the total matched cohort and in the high-risk subgroup. Threshold values of lowest PCO₂ and ECLS probability that were found in the univariate analysis to be associated with lower (+)ECLS mortality, were verified after matching. The primary outcome was compared in the PS matched (+)ECLS and (–)ECLS groups using Kaplan-Meier survival analysis and the log-rank test, and a Cox proportional hazards model to estimate a hazard ratio (HR).

Match 2: High-risk Subgroup

Because variables directly indicative of CDH severity (such as blood gases) are presumably more relevant in the measurement of ECLS benefit than for example demographic characteristics, it was determined that such markers of high risk merit “overweighting” by performing a separate subgroup PS match after controlling for severity before matching.³⁴ Therefore, only high-risk patients (as defined in the unadjusted analysis and verified in Match 1) were included in Match 2. Because there were fewer high-risk patients, the matching algorithm was configured to prioritize the number of matches over match closeness. A more tolerant PS radius of 5% was employed to further increase the number of patients in the matched cohort, again using 1:1 matching in the (+)/(–)ECLS

groups. As with Match 1, the primary outcome was assessed with survival analysis after matching, and the OR and HR for death with ECLS were calculated.

Match 3: Echocardiogram Subgroup

Early cardiac dysfunction is a major and underappreciated determinant of outcomes in CDH.⁶ Thus, to account for this potential confounder, Match 3 included only those ECLS-eligible patients with echocardiogram (echo) data (available in the CDHSG registry since 2015). Additional variables used in PS generation included right or left ventricular dysfunction, and presence of right ventricular systolic pressure $\geq 2/3$ systemic pressure (reflecting moderate-to-severe pulmonary hypertension). As with Match 2, the PS radius was 5% with optimization for number of matches. Again, the primary outcome was assessed using survival analysis, and the OR and HR for death with ECLS were calculated for both the matched cohort and the high-risk subgroup.

Center Effects Analysis

Using all ECLS-eligible patients, CDHSG centers were stratified into low CDH volume (average of <10 CDH cases/yr in the registry) and high CDH volume (average of 10 or more CDH cases/yr), criteria which have been previously described.^{17,29} Centers were further stratified by rate of ECLS use expressed as the fraction (%) of patients who were treated with ECLS out of that center’s CDH population.

Bubble plots were created to model the effect of increasing center ECLS use rate on survival. Robust linear regression was performed, weighted to individual center CDH volume, using Huber M estimation to limit the influence of outliers. Volume-weighted survival was compared between centers with different rates of ECLS use, and between centers for low, medium, and high-risk patients, based on ECLS probability.

Statistical Analysis

Analysis was 2-sided and performed using SAS 9.4, with *P*-values < 0.05 considered significant.

RESULTS

Of 5855 ECLS-eligible CDH patients identified in the registry, 1322 (22.6%) died and 1701 (29.1%) received ECLS (Fig. 1). Clinical characteristics of (+)ECLS patients were indicative of more severe disease; overall mortality in this group was 46.9% (Table 1). ECLS treatment was associated with 6.1 times higher odds of death in the unadjusted cohort.

On subgroup analysis, patients with a lowest PCO₂ of 55 mm Hg or greater during the first day of life ($N = 564$) had lower mortality in the (+)ECLS group [56.9% vs 65.7%, OR 0.69, 95% confidence interval (CI) 0.49–0.9, $P = 0.04$]. Although 55 mm Hg was the observed threshold value for statistical significance, the survival advantage for (+)ECLS patients was more significant in the subgroup with a lowest PCO₂ of 60 mm Hg or greater ($N = 422$, mortality 58.4% vs 76.8%, OR 0.42, 95% CI 0.27–0.66, $P = 0.0001$). Modeled ECLS probability of 80% or greater ($N = 472$) was also found to be a threshold for significantly lower mortality in the (+)ECLS group (57.8% vs 73%, OR 0.59, 95% CI 0.4–0.88, $P = 0.01$).

Match 1: Total Cohort

There were 891 patients in each group after matching, with good covariate balance (Table 1). The OR for death with (+)ECLS was nearly half that of the pre-match (+)ECLS group but remained high. Using the thresholds identified in the pre-match cohort, patients with a lowest PCO₂ of 60 mm Hg or greater ($N = 213$) had lower

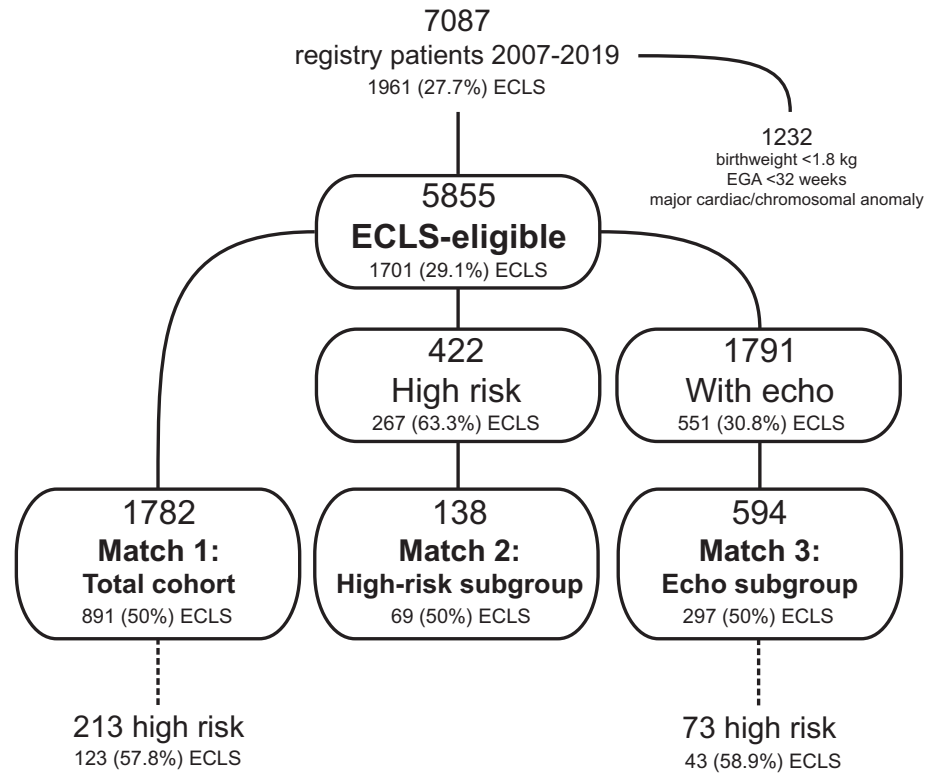


FIGURE 1. Flow chart showing the study cohort and the composition of each of the 3 matches. ECLS rates for each group are also shown. ECLS indicates extracorporeal life support.

TABLE 1. Covariates and Outcomes for all ECLS-eligible Patients Before and After Matching for Propensity Score, Which Was Generated Using the Listed Variables (Match 1)

Variable	Baseline Cohort N = 5855			PS Matched Cohort N = 1782		
	(−)ECLS	(+)ECLS	P	(−)ECLS	(+)ECLS	P
N (%)	4148 (70.9)	1701 (29.1)		891 (50.0)	891 (50.0)	
Male (%)	2458 (59.3)	978 (57.5)	0.21	519 (58.3)	527 (59.2)	0.7
Race/ethnicity						
White (%)	2245 (60.1)	954 (61.7)	<0.0001	565 (63.4)	555 (62.3)	0.97
Black (%)	195 (5.2)	221 (14.3)		94 (10.6)	100 (11.2)	
Hispanic (%)	551 (14.8)	259 (16.7)		172 (19.3)	170 (19.1)	
Asian (%)	611 (16.4)	65 (4.2)		31 (3.5)	33 (3.7)	
Other (%)	95 (2.5)	36 (2.3)		20 (2.2)	25 (2.8)	
Birth weight (kg)	3.04 ± 0.5	3.03 ± 0.5	0.64	3.05 ± 0.6	3.05 ± 0.5	0.87
Prenatal diagnosis (%)	2771 (66.7)	1358 (79.8)	<0.0001	720 (80.8)	710 (79.7)	0.55
High-volume center (%)	2442 (58.8)	1194 (70.2)	<0.0001	593 (66.6)	588 (66.0)	0.8
Inborn (%)	2177 (54.5)	1002 (58.9)	<0.0001	521 (58.5)	517 (58.0)	0.85
EGA birth (wk)	37.8 ± 1.7	37.9 ± 1.6	0.2	37.8 ± 1.9	37.8 ± 1.6	0.79
Left-sided defect	3590 (86.8)	1332 (78.9)	<0.0001	721 (81.3)	721 (81.3)	1
Apgar						
1 min	5.46 ± 2.4	4.08 ± 2.3	<0.0001	4.13 ± 2.2	4.15 ± 2.3	0.84
5 min	7.09 ± 2.0	6.06 ± 2.1	<0.0001	6.23 ± 2.0	6.2 ± 2.1	0.78
Lowest pCO ₂ (mm Hg)	37.5 ± 14.8	46.5 ± 21.4	<0.0001	42.1 ± 18.8	43.6 ± 17.4	0.07
Highest pCO ₂ (mm Hg)	67.0 ± 26.3	89.5 ± 29.2	<0.0001	83.7 ± 30.6	84.0 ± 26.5	0.82
Outcomes						
Death (%)	524 (12.6)	798 (46.9)	<0.0001	194 (21.8)	426 (47.8)	<0.0001
OR death with ECLS (95% CI)		6.1 (5.4–7.0)			3.3 (2.7–4.0)	

High center volume is defined as an average of 10 or more cases of CDH per year submitted to the registry.

Continuous variables given as mean ± standard deviation.

CI indicates confidence interval; EGA, estimated gestational age; OR, odds ratio; pCO₂, partial pressure of carbon dioxide from the postductal arterial blood gas during the first 24 h of life.

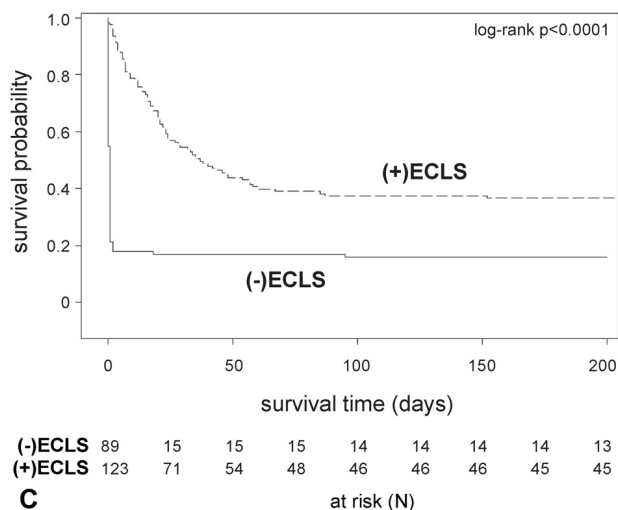
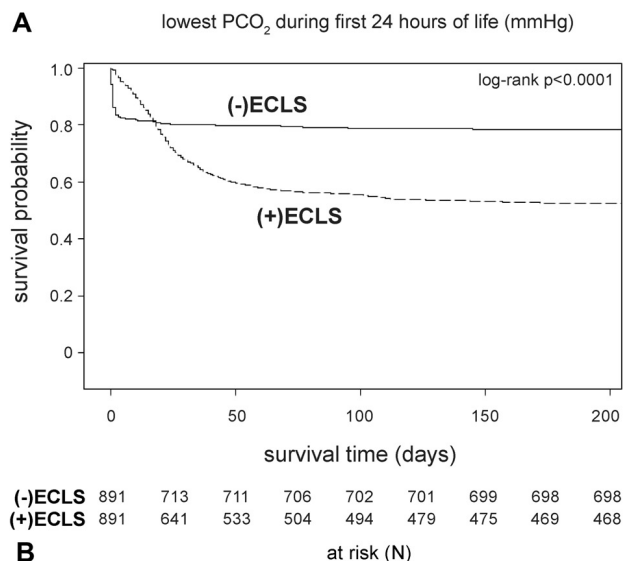
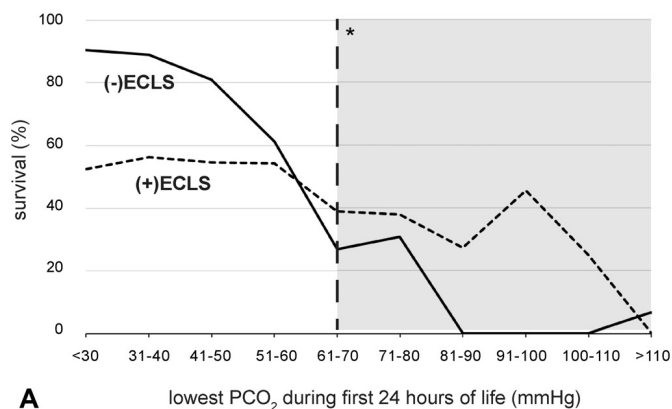


FIGURE 2. A, Survival rates in the (+)ECLS and (-)ECLS groups after PS matching (Match 1), with patients stratified by lowest arterial PCO₂ during the first day of life. Survival was significantly higher with (+)ECLS in patients with a lowest PCO₂ of 60 mm Hg or greater (* $N = 213$, OR for mortality 0.33, 95% CI 0.17–0.65, $P = 0.001$); shaded area designates the clinical region of ECLS survival advantage. B, Kaplan-Meier survival curves comparing 200-d survival in the Match 1 cohort

mortality with (+)ECLS (64.2% vs 84.4%, OR 0.33, 95% CI 0.17–0.65, $P = 0.001$). The clear transition from associated ECLS mortality risk to ECLS survival benefit is graphically represented in Fig. 2A for this clinical threshold. In contrast, an ECLS probability of 80% or greater ($N = 303$) was not associated with a significant mortality difference between the (+)ECLS and (-)ECLS groups (65.8% vs 70.3%, OR 0.81, 95% CI 0.5–1.3, $P = 0.41$).

Survival analysis for the matched cohort comparing (+)ECLS with (-)ECLS showed higher mortality with ECLS over time, using Kaplan-Meier estimates (Fig. 2B, log-rank $P < 0.0001$) and Cox regression (HR 2.3, $P < 0.0001$). Patients with a lowest PCO₂ of 60 mm Hg or greater; however, had significantly lower mortality with ECLS (Fig. 2C, log-rank $P < 0.0001$; HR 0.34, $P < 0.0001$).

Match 2: High-risk Subgroup

Because of the persistent ECLS survival benefit seen after Match 1 for patients with a lowest PCO₂ of 60 mm Hg or greater, this value was used as the cut-off to define the “high risk” threshold for Match 2. After matching using just this subgroup of patients ($N = 422$, Fig. 1), there were 69 in each of the (+) and (-)ECLS groups with good covariate balance (Table 2). Overall mortality was lower with (+)ECLS (60.9% vs 79.7%, OR 0.4, 95% CI 0.19–0.85, $P = 0.016$). Kaplan-Meier analysis showed lower mortality with (+)ECLS (log-rank $P < 0.0001$) and the HR was 0.41 using Cox regression ($P < 0.0001$).

Match 3: Echocardiogram Subgroup

There were 1791 patients with echo data. Although the clinical characteristics of these patients were similar to the original cohort, the OR for death with (+)ECLS was higher. Patients in the (+)ECLS group had much higher rates of right ventricular dysfunction, left ventricular dysfunction and pulmonary hypertension. After matching, there were 297 patients in each group, with acceptable covariate balance (Table 3). As with Match 1, the OR for death with (+)ECLS was much lower than the pre-match OR, though (+)ECLS mortality was essentially unchanged after matching. Survival analysis was similar to Match 1 (log-rank $P < 0.0001$); Cox regression for the Match 3 cohort showed that (+)ECLS had a HR of 2.45 ($P < 0.0001$).

Matched patients with a lowest PCO₂ of 60 mm Hg or greater ($N = 73$) had significantly lower mortality in the (+)ECLS group (48.8% vs 76.7%, OR 0.29, 95% CI 0.1–0.82, $P = 0.017$). Kaplan-Meier analysis again showed lower mortality with (+)ECLS (log-rank $P < 0.0001$), with a HR by Cox regression of 0.32 ($P = 0.0003$).

Center Effects Analysis

There were 102 individual centers represented in the data set. An ECLS probability of $\leq 20\%$ ($N = 1252$) defined “low risk” patients. Mortality in this group was 4.8%. Patients with an ECLS probability of 20%–80% ($N = 2109$) had 25.1% mortality and this group was defined as “medium risk.” An ECLS probability of 80%

between all patients in the (+)ECLS and (-)ECLS groups, and C, between patients with lowest PCO₂ of 60 mm Hg or greater during the first 24 h of life in the (+)ECLS and (-)ECLS groups. Patients were censored at time of discharge/transfer, and time to event (death) was measured from birth in days. Because >99% of deaths occurred before 200 days of life, a survival time cut-off of 200 d was used to simplify Kaplan-Meier graphical representation. At-risk patients (N) shown at bottom of each curve for each time point. CI indicates confidence interval; ECLS, extracorporeal life support; OR, odds ratio; PCO₂, partial pressure of carbon dioxide.

TABLE 2. Covariates and Outcomes for ECLS-eligible High-risk Patients Before and After Matching for Propensity Score Which Was Generated Using the Listed Variables (Match 2). A "High Risk" Patient is Defined Here as a Lowest pCO₂ of 60 mm Hg or Greater During the First 24 h of Life

Variable	High-risk Subgroup N = 422			PS Matched High-risk Subgroup N = 138		
	(-)ECLS	(+)ECLS	P	(-)ECLS	(+)ECLS	P
N (%)	155 (36.7)	267 (63.3)		69 (50.0)	69 (50.0)	
Male (%)	81 (52.3)	156 (58.7)	0.2	37 (53.6)	36 (53.6)	1
Race/ethnicity						
White (%)	74 (51.8)	161 (64.4)	<0.0001	43 (62.3)	44 (63.8)	0.98
Black (%)	16 (11.2)	50 (20.0)		11 (15.9)	9 (13.0)	
Hispanic (%)	18 (12.6)	30 (12.0)		11 (15.9)	11 (15.9)	
Asian (%)	31 (21.7)	4 (1.6)		2 (2.9)	2 (2.9)	
Other (%)	3 (2.1)	4 (1.6)		2 (2.9)	3 (4.4)	
Birth weight (kg)	2.8 ± 0.6	3.04 ± 0.5	<0.0001	2.88 ± 0.6	2.9 ± 0.5	0.85
Prenatal diagnosis (%)	128 (82.6)	226 (84.6)	0.39	55 (79.7)	55 (79.7)	1
High-volume center (%)	88 (56.8)	192 (71.9)	0.002	51 (73.9)	50 (72.5)	0.85
Inborn (%)	109 (70.3)	173 (64.8)	0.24	44 (63.8)	37 (53.6)	0.23
EGA birth (wk)	36.8 ± 2.1	37.6 ± 1.6	<0.0001	37.1 ± 1.9	37.1 ± 1.8	1
Left-sided defect	114 (77.0)	197 (74.3)	0.54	48 (72.7)	56 (83.6)	0.13
Apgar						
1 min	3.49 ± 2.3	3.64 ± 2.2	0.53	3.87 ± 2.3	3.81 ± 2.3	0.88
5 min	4.85 ± 2.6	5.69 ± 2.0	0.0003	5.29 ± 2.1	5.52 ± 2.1	0.52
Lowest pCO ₂ (mm Hg)	85.0 ± 29.1	81.9 ± 21.8	0.21	84.4 ± 28.7	80.4 ± 19.8	0.34
Highest pCO ₂ (mm Hg)	121.1 ± 33.0	111.5 ± 21.1	0.002	120.1 ± 32.6	117.8 ± 25.1	0.65
Outcomes						
Death (%)	119 (76.8)	156 (58.4)	0.0001	55 (79.7)	42 (60.9)	0.016
OR death with ECLS		0.42 (0.27–0.66)			0.4 (0.19–0.85)	

High center volume is defined as an average of 10 or more cases of CDH per year submitted to the registry. Continuous variables given as mean ± standard deviation.
 CI indicates confidence interval; EGA, estimated gestational age; OR, odds ratio; pCO₂, partial pressure of carbon dioxide from the postductal arterial blood gas during the first 24 h of life.

TABLE 3. Covariates and Outcomes for ECLS-eligible Patients With First-day Echocardiogram Data, Before and After Matching for Propensity Score Generated Using the Listed Variables (Match 3)

Variable	Patients With Echocardiogram N = 1791			PS Matched Echocardiogram Cohort N = 594		
	(-)ECLS	(+)ECLS	P	(-)ECLS	(+)ECLS	P
N (%)	1240 (69.2)	551 (30.8)		297 (50.0)	297 (50.0)	
Male (%)	720 (58.1)	304 (55.2)	0.25	176 (59.3)	164 (55.2)	0.32
Race/ethnicity						
White (%)	660 (54.7)	321 (60.0)	<0.0001	179 (60.3)	176 (59.3)	0.99
Black (%)	66 (5.5)	70 (13.1)		33 (11.1)	37 (12.5)	
Hispanic (%)	239 (19.8)	107 (20.0)		66 (22.2)	65 (21.9)	
Asian (%)	190 (15.8)	21 (3.9)		10 (3.4)	10 (3.4)	
Other (%)	36 (3.0)	14 (2.6)		6 (2.0)	7 (2.4)	
Birth weight (kg)	3.03 ± 0.5	3.01 ± 0.5	0.35	3.02 ± 0.5	3.01 ± 0.5	0.89
Prenatal diagnosis (%)	912 (73.6)	489 (88.8)	<0.0001	265 (89.2)	261 (87.9)	0.6
High-volume center (%)	745 (60.1)	397 (72.1)	<0.0001	219 (73.7)	209 (70.4)	0.36
Inborn (%)	751 (60.6)	380 (69.0)	0.0007	200 (67.3)	196 (66.0)	0.73
EGA birth (wk)	37.8 ± 1.7	37.8 ± 1.6	0.46	37.8 ± 1.8	37.7 ± 1.7	0.44
Left-sided defect	1082 (87.5)	425 (77.7)	<0.0001	244 (82.4)	238 (80.1)	0.47
Apgar						
1 min	5.44 ± 2.4	3.87 ± 2.3	<0.0001	4.27 ± 2.3	4.21 ± 2.4	0.74 0.97
5 min	7.07 ± 2.0	5.89 ± 2.1	<0.0001	6.18 ± 2.1	6.18 ± 2.0	
Lowest pCO ₂ (mm Hg)	37.3 ± 13.6	46.8 ± 21.3	<0.0001	41.5 ± 18.9	44.7 ± 19.9	0.05
Highest pCO ₂ (mm Hg)	66.0 ± 25.8	88.4 ± 27.3	<0.0001	79.0 ± 30.2	83.2 ± 25.9	0.07
Right ventricular dysfunction	242 (19.5)	238 (43.2)	<0.0001	106 (35.7)	108 (36.4)	0.75
Left ventricular dysfunction	141 (11.4)	196 (35.6)	<0.0001	70 (23.6)	75 (25.3)	0.89
Moderate-severe PHTN	744 (60.0)	471 (85.5)	<0.0001	259 (87.2)	257 (86.5)	0.92
Outcomes						
Death (%)	131 (10.6)	275 (49.9)	<0.0001	73 (23.3)	148 (47.1)	<0.0001
OR death with ECLS (95% CI)		8.4 (6.6–10.8)			3.5 (2.5–5.1)	

High center volume is defined as an average of 10 or more cases of CDH per year submitted to the registry. Continuous variables given as mean ± standard deviation.
 CI indicates confidence interval; EGA, estimated gestational age; OR, odds ratio; pCO₂, partial pressure of carbon dioxide from the postductal arterial blood gas during the first 24 h of life.

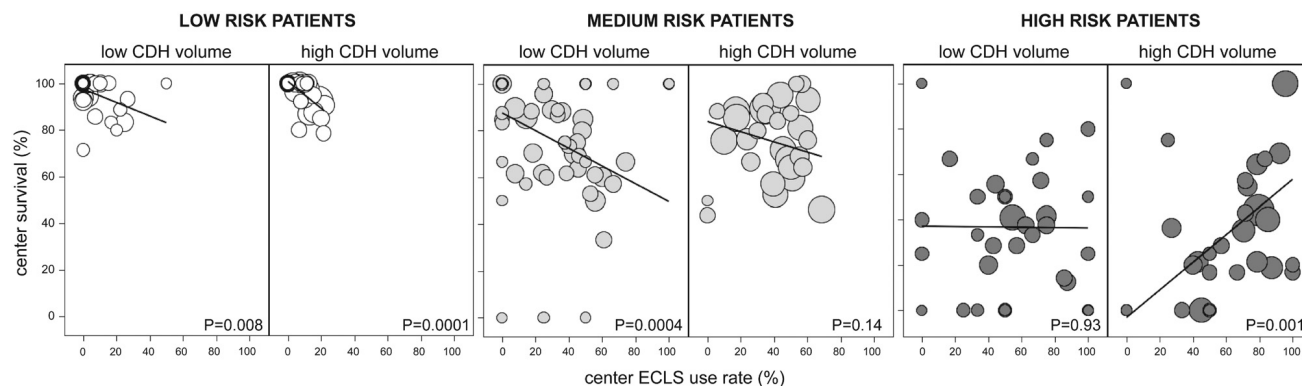


FIGURE 3. Bubble plots comparing low CDH volume to high CDH volume centers (average of <10 vs 10 or more cases per year, total of 102 centers), plotting % center overall survival against % center ECLS use rate for 3 groups: low-risk (N = 1252), medium-risk (N = 2109), and high-risk (N = 472) ECLS-eligible patients (<20%, 20–80%, and ≥80% predicted probability of ECLS use). Lines and P-values represent robust linear regression weighted to center CDH volume for each risk tier. Each bubble represents a single-center, and bubble size is directly proportional to the number of patients (volume) from each center in that risk category. CDH indicates congenital diaphragmatic hernia; ECLS, extracorporeal life support.

or greater (N = 472) defined “high risk.” Mortality in the high-risk group was 62.1%.

With patients stratified into these 3 risk tiers, center survival was plotted against center ECLS use rate for high and low-volume centers (Fig. 3). For low-risk patients, center survival significantly decreased with increasing ECLS use rates at both low and high-volume centers. For medium-risk patients, a significant decrease in survival with increasing ECLS use rates was seen only at low-volume centers. For high-risk patients at low-volume centers, there was no significant change in survival with increasing ECLS use rates, and survival rates were highly variable (R^2 0.0003). In contrast, at high-volume centers, there was a clearly significant increase in survival seen for high-risk patients with increasing center ECLS use rates, and there was much less survival variance (R^2 0.25).

DISCUSSION

Despite conflicting evidence regarding the efficacy of ECLS in the neonate with CDH, nearly a third will receive ECLS treatment during their course of care, with approximately half of those patients surviving. The degree of potential survival benefit from ECLS for CDH patients has never been accurately quantified, and it has not been clear to which patients or at which centers any such benefit is likely to apply. Using a PS approach to reduce the effects of confounding, this study identified both excess overall mortality associated with ECLS use, and specific thresholds for disease severity above which patients had significantly lower mortality with ECLS use. Patients with a lowest achieved PCO₂ of 60 mm Hg or greater during the first day of life were more than twice as likely to survive with ECLS use than without. Although the degree to which ECLS offers a survival advantage is dependent on the definition of “high risk,” the advantage for ECLS in the sickest patients was found to persist using strict statistical modeling taking into consideration all available and/or known potential confounders. This benefit was not apparent in other patients. In fact, we found that the use of ECLS in all but high-risk patients was associated with much higher mortality – underlining the importance of reliable measures of disease severity, optimal critical care management, and standardized care pathways.

This investigation supports previous work suggesting a survival benefit with ECLS treatment.^{13,35,36} In 1999, the CDHSG reported a survival advantage with ECLS use only for patients with a predicted mortality rate in excess of 80% (N = 30), using the 5

minute Apgar score and birth weight to risk-stratify those patients.¹¹ In 1996, a small randomized trial with a CDH subgroup analysis (N = 35) showed no long-term survival advantage with ECLS (survival 22% with ECLS vs 0% without).¹⁰ Two small prospective trials with CDH subgroups (N = 2 for both) are uninterpretable given the known spectrum of severity for the disease.^{37,38} It is unlikely that a sufficiently powered randomized trial will be done to definitively evaluate this question, and thus the present study may represent optimal evidence justifying the use of this complex, invasive, and expensive modality.

Both CDH and ECLS center volume have been shown to affect CDH outcomes.^{16,17} In our study, because the propensity matching did not completely adjust for center-level characteristics, a risk-stratified center effects analysis was performed to more clearly demonstrate the effects of both center ECLS use rates and center CDH volume on survival in different risk categories of CDH patients. Importantly, the survival advantage with ECLS was only clearly demonstrable for high-risk patients at centers with high CDH volume (10 or more cases per year) and high rates of ECLS treatment (which implies commensurate extensive experience with ECLS). At low-volume centers, higher rates of ECLS use were associated with equivalent or worse survival for patients of all risk types. Disparate management and ECLS selection criteria likely contribute to observed survival variability at less experienced centers. Moreover, this analysis confirms previous work highlighting volume-outcome and volume-variability relationships.^{29,39}

Higher center-level mortality seen with increasing center ECLS rates in low-risk patients may reflect occult severity in CDH patients who receive ECLS despite initially favorable postnatal clinical parameters. Recent evidence identified early left ventricular dysfunction and pulmonary hypertension as findings associated with adverse outcomes among patients with otherwise seemingly low-risk CDH.⁶ It is also possible that improper patient selection or excess pre-ECLS lung injury is occurring at some centers, leading to poor outcomes with unanticipated ECLS used as salvage therapy. Irrespective of the fundamental reason, the particularly low survival seen with ECLS use in otherwise low-risk patients emphasizes the importance of safe, high-quality ECLS and proper patient selection for ECLS, and reinforces the risk associated with ECLS treatment.

These results allude to the potential value of standardized clinical practice guidelines to improve future outcomes in CDH.

Variability in survival of CDH patients between centers is apparent and is likely related to factors including patient volume, ECLS experience, and patient selection.⁴⁰ Several, multi-institutional, multi-disciplinary consensus guidelines exist regarding CDH management.^{3,41} Implementation of standard criteria surrounding ECLS use could allow aligned CDH care across centers and provide a foundation for large, prospective studies examining elements of optimal management. The present study has identified clinical thresholds for ECLS benefit and will assist in the development of a set of ECLS initiation and management guidelines for CDH.

This study is limited by its retrospective nature, and by the fact that PS matching can only reduce bias in measured characteristics; other/unknown covariates are unaccounted for and maybe confounding the data. Despite its limitations, PS matching is a useful tool in the study of complex diseases such as CDH where randomized trials are extremely difficult to perform.^{42,43} Another limitation of our analysis is that mortality is only 1 measure of outcome with ECLS use in CDH; other important considerations such as resource utilization, value, neurodevelopmental outcomes, and quality of life should be assessed in future CDH studies.

In summary, this study demonstrates that ECLS affords a modest survival advantage for high-risk CDH patients but is associated with worse survival in low-to-moderate-risk patients. The advantage for high-risk patients is heavily influenced by center CDH volume and ECLS experience. Identified clinical thresholds have utility in the development of future prospective studies and in the construction of CDH clinical practice guidelines, which may help reduce treatment variability and improve overall survival.

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